

Final Armageddon Practical-1(Puppet): Configuration-Driven Web Server Deployment with Puppet, Hiera, Facter, EPP Templates and Apache VirtualHosts

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/calp$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ find .
.
./environment.conf
./data
./data/nodes
./data/nodes/puppet-agent.yaml
./data/common.yaml
./hiera.yaml
./manifests
./manifests/site.pp
./modules
./modules/calp
./modules/calp/templates
./modules/calp/templates/index.html.epp
./modules/calp/templates/employee_portal.conf.epp
./modules/calp/manifests
./modules/calp/manifests/config.pp
./modules/calp/manifests/service.pp
./modules/calp/manifests/init.pp
./modules/calp/manifests/install.pp
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
GNU nano 6.2 data/nodes/puppet-agent.yaml
calp::config::portal_name: "Company Employee Information Portal"
calp::config::department: "Information Technology"
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
GNU nano 6.2 data/common.yaml
calp::config::company_msg: "This portal is deployed automatically by Vishnu and"
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd manifests
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ sudo
o cat site.pp
node puppet-agent{
  include calp
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ cd
..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd modules/ca
lp/manifests
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/calp/m
anifests$ sudo cat init.pp
class calp{

  contain calp::install
  contain calp::config
  contain calp::service

  Class['calp::install']->Class['calp::config']->Class['calp::service']

}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/calp/m
anifests$ sudo cat config.pp
class calp::config(

```

```

}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/calp/m
anifests$ sudo cat config.pp
class calp::config(
  String $portal_name,
  String $department,
  String $company_msg
){

  file{'/var/www/employee_portal':
    ensure=>directory,
    owner=>'www-data',
    group=>'www-data',
    mode=>'0755'
  }

  file{'/var/www/employee_portal/index.html':
    ensure=>file,
    owner=>'www-data',
    group=>'www-data',
    mode=>'0644',
    content=>epp('calp/index.html.epp',
    {

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
group=>'www-data',
mode=>'0644',
content=>epp('ca1p/index.html.epp',
{
  'portal_name'=>$portal_name,
  'department'=>$department,
  'company_msg'=>$company_msg
}),
notify=>Service['apache2'],
require=>File['/var/www/employee_portal']
}

file{'/etc/apache2/sites-available/employee_portal.conf':
  ensure=>file,
  owner=>'root',
  group=>'root',
  mode=>'0644',
  content=>epp('ca1p/employee_portal.conf.epp'),
  notify=>Service['apache2']
}

exec{'disable-default-site':
  command=>'/usr/sbin/a2dissite 000-default.conf',
  onlyif=>'/usr/sbin/a2query -s 000-default.conf',
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
mode=>'0644',
content=>epp('ca1p/employee_portal.conf.epp'),
notify=>Service['apache2']
}

exec{'disable-default-site':
  command=>'/usr/sbin/a2dissite 000-default.conf',
  onlyif=>'/usr/sbin/a2query -s 000-default.conf',
  notify=>Service['apache2']
}

exec{'enable-actual-site':
  command=>'/usr/sbin/a2ensite employee_portal.conf',
  unless=>'/usr/sbin/a2query -s employee_portal.conf',
  require=>File['/etc/apache2/sites-available/employee_portal.conf'],
  notify=>Service['apache2']
}
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/ca1p/m
anifests$ sudo cat service.pp
class ca1p::service{

  service{'apache2':
    ensure=>running,
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
service{'apache2':
  ensure=>running,
  enable=>true
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/ca1p/m
anifests$ sudo cat install.pp
class calp::install{

  package{'apache2':
    ensure=>installed
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/ca1p/m
anifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/ca1p$
cd templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/ca1p/t
emplates$ sudo cat index.html.epp
<h1> Welcome to the Company Employee Portal </h1>

<h2> <%= $portal_name %> </h2>
<h2> Department: <%= $department %> </h2>
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
templates$ sudo cat index.html.epp
<h1> Welcome to the Company Employee Portal </h1>

<h2> <%= $portal_name %> </h2>
<h2> Department: <%= $department %> </h2>

<h3> Hostname: <%= $facts['networking']['hostname'] %> </h3>
<h3> IP Address: <%= $facts['networking']['ip'] %> </h3>
<h3> Operating System: <%= $facts['os']['name'] %> </h3>

<h3> <%= $company_msg %> </h3>
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/ca1p/t
emplates$ sudo cat employee_portal.conf.epp
<VirtualHost *:80>

  DocumentRoot /var/www/employee_portal

  <Directory /var/www/employee_portal>
    Require all granted
  </Directory>
</VirtualHost>
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/ca1p/t
emplates$
```

Agent:

```
ubuntu@puppet-agent: ~
Notice: Requesting catalog from puppet-master:8140 (172.31.9.93)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent
Info: Applying configuration version '1787910316'
Notice: /Stage[main]/Ca1p::Config/File[/var/www/employee_portal]/ensure: create
Notice: /Stage[main]/Ca1p::Config/File[/var/www/employee_portal/index.html]/ens
re: defined content as '{sha256}b6a7808e5548ae8c3545d9e7ee24e093f1325c0b8dd3748
7b8fff1dce28b573'
Info: /Stage[main]/Ca1p::Config/File[/var/www/employee_portal/index.html]: Sched
uling refresh of Service[apache2]
Notice: /Stage[main]/Ca1p::Config/File[/etc/apache2/sites-available/employee_po
tal.conf]/ensure: defined content as '{sha256}0adc32307086bbcd2ccf418520d02963f
c99efdd5f06db04c05098c5e4abf8e'
Info: /Stage[main]/Ca1p::Config/File[/etc/apache2/sites-available/employee_port
l.conf]: Scheduling refresh of Service[apache2]
Notice: /Stage[main]/Ca1p::Config/Exec[disable-default-site]/returns: executed s
uccessfully
Info: /Stage[main]/Ca1p::Config/Exec[disable-default-site]: Scheduling refresh o
f Service[apache2]
Notice: /Stage[main]/Ca1p::Config/Exec[enable-actual-site]/returns: executed su
ccessfully
Info: /Stage[main]/Ca1p::Config/Exec[enable-actual-site]: Scheduling refresh of
Service[apache2]
Notice: /Stage[main]/Ca1p::Service/Service[apache2]: Triggered 'refresh' from 4
```

```
ubuntu@puppet-agent: ~
re: defined content as '{sha256}b6a7808e5548ae8c3545d9e7ee24e093f1325c0b8dd37484
7b8fff1dce28b573'
Info: /Stage[main]/Ca1p::Config/File[/var/www/employee_portal/index.html]: Sched
uling refresh of Service[apache2]
Notice: /Stage[main]/Ca1p::Config/File[/etc/apache2/sites-available/employee_por
tal.conf]/ensure: defined content as '{sha256}0adc32307086bbcd2ccf418520d02963f2
c99efdd5f06db04c05098c5e4abf8e'
Info: /Stage[main]/Ca1p::Config/File[/etc/apache2/sites-available/employee_porta
l.conf]: Scheduling refresh of Service[apache2]
Notice: /Stage[main]/Ca1p::Config/Exec[disable-default-site]/returns: executed s
uccessfully
Info: /Stage[main]/Ca1p::Config/Exec[disable-default-site]: Scheduling refresh o
f Service[apache2]
Notice: /Stage[main]/Ca1p::Config/Exec[enable-actual-site]/returns: executed suc
cessfully
Info: /Stage[main]/Ca1p::Config/Exec[enable-actual-site]: Scheduling refresh of
Service[apache2]
Notice: /Stage[main]/Ca1p::Service/Service[apache2]: Triggered 'refresh' from 4
events
Notice: Applied catalog in 0.78 seconds
ubuntu@puppet-agent:~$ ls -l /etc/apache2/sites-enabled/
total 0
lrwxrwxrwx 1 root root 39 Aug 28 09:45 employee_portal.conf -> ../sites-availabl
e/employee_portal.conf
```



Welcome to the Company Employee Portal

Company Employee Information Portal

Department: Information Technology

Hostname: puppet-agent

IP Address: 172.31.1.8

Operating System: Ubuntu

This portal is deployed automatically by Vishnu and Puppet

```
ubuntu@puppet-agent: ~  
Active: active (running) since Fri 2026-08-28 09:51:49 UTC; 12min ago  
Docs: https://httpd.apache.org/docs/2.4/  
Process: 3078 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)  
Main PID: 3083 (apache2)  
Tasks: 55 (limit: 1018)  
Memory: 5.6M  
CPU: 69ms  
CGroup: /system.slice/apache2.service  
├─3083 /usr/sbin/apache2 -k start  
├─3084 /usr/sbin/apache2 -k start  
└─3085 /usr/sbin/apache2 -k start  
Aug 28 09:51:49 puppet-agent systemd[1]: Starting The Apache HTTP Server...  
Aug 28 09:51:49 puppet-agent systemd[1]: Started The Apache HTTP Server.  
ubuntu@puppet-agent:~$ sudo ls -l /etc/apache2/sites-available  
total 16  
-rw-r--r-- 1 root root 1332 Mar 18 2024 000-default.conf  
-rw-r--r-- 1 root root 6338 Mar 18 2024 default-ssl.conf  
-rw-r--r-- 1 root root 157 Aug 28 09:45 employee_portal.conf  
ubuntu@puppet-agent:~$ sudo ls -l /etc/apache2/sites-enabled  
total 0  
lrwxrwxrwx 1 root root 39 Aug 28 09:45 employee_portal.conf -> ../sites-available/employee_portal.conf  
ubuntu@puppet-agent:~$ |
```